



Genome-Wide Identification and In Silico Analysis of Annexins in Chickpea (*Cicer arietinum* L.)

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Abstract

Annexins are a ubiquitous, evolutionarily conserved group of Ca^{2+} -dependent phospholipid-binding proteins. They are a family of less numerous and more varied proteins that form a unique monophyletic group. They play an important role in various abiotic and biotic stress responses through Ca^{2+} -mediated signaling. Chickpea (*Cicer arietinum* L.) is one of the most widely grown legume crops in the world. In recent years, intensive research has been carried out to identify and elucidate genes and molecular pathways that control stress responses in plants. The availability of the chickpea genome has hastened the functional genomics of chickpea. In the current study, we attempted Genome-wide identification and in silico analysis of Annexins in chickpea. Thirteen annexin sequences have been identified in the chickpea genome. Four conserved annexin domains were found in ten annexin members, while three annexins CaAnn5, CaAnn12, and CaAnn13, showed three, two, and one conserved domain, respectively. The gene structure analysis showed the presence of multiple exons in all thirteen annexins. Most Annexin genes are composed of 3–5 introns. Their chromosomal locations showed that out of thirteen genes, ten could be mapped on four chromosomes. Three genes were placed on the scaffold regions. The promoter sequence analysis of all thirteen annexins showed the presence of various elements related to growth and development and response to different phytohormones and abiotic stress. The gene expression data of different annexins in various tissues like leaf, shoot, root, flower bud, and young pod showed their differential expression. Analysis of expression data of roots in drought stress showed their differential expression with the different stages of plant growth. Overall, the current findings show the possible role of CaAnns in different stages of plant growth and development in normal and stressful conditions. Moreover, these findings will be helpful in the further characterization of CaAnn genes and their promoters.

Keywords Annexins · Chickpea · Genome · Gene expression · Promoter · Stress

Introduction

For plants, it is difficult to escape from harsh environmental conditions due to their sessile nature. Plant's growth and development are hampered by rapid changes in their environment caused by various biotic and abiotic stressors, which reduces their productivity (Chaudhry and Sidhu 2022). Plants have a variety of physiological and molecular acclimation processes that are employed to cope with different environmental stresses (Mittler and Blumwald 2015; Büyüç et al. 2016). The most common abiotic stresses include drought, salinity, high and low temperatures, and heavy metal toxicity (Zeng et al. 2015). Abiotic stress could ultimately lead to the cell death and loss of cell homeostasis (Wang et al. 2003; Cramer et al. 2011). Improvement of crop resistance for stress becomes a necessity because it is very challenging to physically eliminate abiotic stress. Intensive research has been carried out in many crop plants to identify the genes and molecular mechanisms for the management of stress (Peleg et al. 2011; Gill et al. 2015; Yadav et al. 2016; Ahmed et al. 2017; Joshi et al. 2020). Different transcription factors (TFs) and their DNA binding sites, the cis-acting elements, function as molecular switches for stress-responsive gene expression within the regulatory networks that control the signal transduction from stress signal perception to stress-responsive gene expression, enabling plants to adapt to the negative stressor more effectively (Van et al. 2014; Bhattacharjee et al. 2015; Sun et al. 2015; Die et al. 2018). So, there is a need for the identification of different molecular components that regulate stress tolerance in plants and can be used as markers or candidates for crop improvement (Meena et al. 2015a).

Almost every biological process in eukaryotic organisms is regulated by Ca^{2+} (Batišć and Kudla 2009). Calcium has been widely considered as a ubiquitous secondary messenger of environmentally and physiologically triggered signaling pathways in plants. It regulates various cellular processes in plants as an essential second messenger (Dodd et al. 2010; Chen et al. 2013). Ca^{2+} transmits various external and internal signals, and therefore, it is very important for the plant's stress response process (Tuteja and Mahajan 2007; Ahmed et al. 2018). Specific temporal changes in the cytosolic Ca^{2+} concentration $[\text{Ca}^{2+}]_{\text{cyt}}$ are induced by both internal and environmental stimuli, including developmental, light, and stresses. The magnitude and kinetics of calcium ion concentration known as the calcium signature which may affect the particular response (Bose et al. 2011; Gilroy et al. 2014). A wide variety of Ca^{2+} binding proteins decode and communicate these calcium signatures, relaying this information into downstream responses (Tuteja and Mahajan 2007; Meena et al. 2015b; Yadav et al. 2018). Major Ca^{2+} sensors that have been characterized and identified in plants include calcineurin B-like proteins (CBLs), calmodulin (CaM), Ca^{2+} -dependent protein kinases (CDPKs), CBL-interacting protein kinases (CIPKs), and CaM-like proteins (CMLs) (Hashimoto and Kudla 2011). Annexins are also among them and make a Ca^{2+} sensor family in the plant kingdom (Clark et al. 2012; Davies 2014; Yadav et al. 2018).

Plant Annexins are ubiquitous, an evolutionarily conserved group of Ca^{2+} -dependent phospholipid-binding proteins (Moss and Morgan 2004;

Konopka-Postupolska et al. 2011; He et al. 2020). Annexin was first discovered in plant i.e., in tomato plants by (Boustead et al. 1989). Plant annexins are a family of less numerous and more varied proteins that form a unique monophyletic group (Jami et al. 2012a). The basic plant annexin is made up of annexin domains. Each domain comprising of five α helices, and is approximately 70 amino acids long and they contain conserved endonexin sequence (K-G-X-G-T- {38}- D/E) which can bind Ca^{2+} (Mortimer et al. 2008; Konopka-Postupolska et al. 2011). Plant annexins are primarily responsible for membrane binding, membrane trafficking, controlling the growth and development of plants, and preserving calcium homeostasis (Mortimer et al. 2008; Divya et al. 2010; Clark et al. 2012). Annexins are transcriptionally regulated in response to a variety of abiotic stresses (Jami et al. 2010; Dalal et al. 2014; Yadav et al. 2015). Plant annexins are expressed in numerous tissues at various stages of development and are involved in a variety of aspects of plant growth and development (Que et al. 2023). Cellular functions like growth, differentiation, secretion, and repair depend on their participation in membrane organisation, vesicle trafficking, and membrane-related signal transduction (Kodavali et al. 2013). They are thus connected to controlling membrane dynamics, $[\text{Ca}^{2+}]_{\text{cyt}}$ sensing and signaling, $[\text{Ca}^{2+}]_{\text{cyt}}$ dynamics and cytoskeletal responses, and influencing stress responses (Konopka-Postupolska et al. 2009; Laohavisit et al. 2010; Jami et al. 2012a; Boyidi et al. 2021). Initially, annexins were thought to be Ca^{2+} mediated, membrane lipid-associated proteins, but the discovery of Ca^{2+} independent membrane interaction and the discovery of additional conserved structural motifs, distinct functional domains, distinctive amino-termini, and various forms of architecture broke this assumption and proposed integrated roles and mechanisms for individual annexin functions (Fernandez et al. 2017; Yadav et al. 2018).

Chickpea is the second most widely grown legume crop after soybean, accounting for a substantial proportion of human dietary nitrogen intake and playing a crucial role in food security in developing countries (Jukanti et al. 2012). It ranks third in the world for food legume crop production (Ramalingam et al. 2015). A complete genome sequence (738 Mb) and transcriptome assembly are available for the diploid genome ($2x = 2n = 16$) of the chickpea (Varshney et al. 2013; Verma et al. 2015b). Importantly, India, the world's top producer of chickpeas, contributes almost 70% (5.9 million tons (mt) yearly) to the global production of chickpeas (Varshney et al. 2009). There are two main types of chickpeas: small-seeded desi and large-seeded kabuli (van der Maesen 1987). Unfortunately, the productivity of the chickpea is significantly reduced as a result of the difficult and constantly changing environmental conditions in the semi-arid tropics, where it is primarily grown (Kumar and Abbo 2001). Abiotic stress factors may exceed 6.4 million tons of the yearly global yield losses of chickpeas (Jha et al. 2014). The flower set, pollen viability, pod set, and retention are all negatively affected by abiotic stresses. A negative effect on them severely reduces chickpea production as these developmental phases affect the number of seeds in a chickpea (Maphosa et al. 2020). Significant efforts were recently made to develop the chickpea genome resources due to its high nutritional value and commercial significance (Jain et al. 2013; Varshney et al. 2013; Parween et al. 2015). Thus, it is crucially necessary to identify key stress-related genes and use

them in the breeding program to produce stress-tolerant chickpea varieties (Garg et al. 2011; Singh et al. 2013; Deepika et al. 2022; Kudapa et al. 2023).

New proteins and genes have been identified in recent years using computational tools for genome-wide in silico research. Whole genome sequences of various plant species, ranging from unicellular algae to multicellular plants, have recently been available in public databases, providing the opportunity for extensive molecular, evolutionary, and functional insights about annexin gene families (Jami et al. 2012b). In this study, we performed genome-wide identification of the Annexin family in *Cicer arietinum* (*C. arietinum*) Desi-type chickpea (ICC4958). Although the Desi-type chickpea (ICC 4958) genome sequence was available for almost ten years, there is little information about the annexin gene family in chickpeas. In the present investigation, a total of thirteen annexin sequences were identified in Desi-type chickpea (ICC 4958) genome. The evolutionary relatedness of these thirteen chickpea annexins was examined with their orthologs of other species. Moreover, their chromosomal positions, presence of different motifs, gene structure, and conserved domains were discussed. Based on the available transcriptome data for the Desi-type chickpea (ICC 4958), annexin expression in different organs and under different stress conditions at various stages of its life cycle was also analyzed in the current study.

Materials and Methods

Sequence Retrieval of *C. arietinum* Annexins (CaAnns)

To identify the genes that encode for annexin, the chickpea genome at the pulse crop database (PCD) was examined. Their respective mRNA, CDS, and protein sequences were also retrieved from the PCD. For the identification of annexin gene members, the *Medicago truncatula* (*M. truncatula*) annexin sequence was used. The *M. truncatula* protein sequence was downloaded from the phytozome v13 database (<http://www.phytozome.net/>). This *M. truncatula* protein sequence was used as a query in protein BLAST (BLASTP) searches for the identification of annexin gene members of small-seeded desi-type chickpea ICC4958 cultivar in pulse crop database (<https://www.pulsedb.org/>). Based on a BLASTP search with an expected *e* value of $1e-4$, the genes with more than 100 BLAST score values were chosen for further analysis (Yadav et al. 2015). The identified annexin sequences were used as queries to re-confirm the protein sequences in different web servers like in the legume information system (LIS) (<https://www.legumeinfo.org/>) and Phytozome (Goodstein et al. 2012). The re-annotation of identified annexin sequences was carried out.

Identification of Conserved Motifs

The Multiple EM for Motif Elicitation (MEME) web server (MEME—Submission form (meme-suite.org)) was used to analyze all the 13 annexin protein sequences from the desi-type chickpea for the identification of conserved protein motifs. The

option Any number of repetitions (anr) in MEME was used to determine how the motifs were distributed among the sequences (Bailey and Elkan 1994). The number of distinct motifs was set to 20. Along with other parameters, default values were generated with the minimum and maximum sizes of 6 and 50 amino acids, respectively. Functions of motifs were analyzed using InterProScan (Paysan-Lafosse et al. 2023) and Pfam (Mistry et al. 2021).

Prediction of Gene Structure and Structural Domains

The annexin gene sequences and their CDS sequences were downloaded from PCD to analyze the gene structure of chickpea Annexins. These sequences were uploaded to the Gene Structure Display Server 2.0 (Hu et al. 2015) to generate the gene structure diagram. The conserved domains were determined using the SMART database (<http://smart.embl-heidelberg.de/>) which provides information about the positions of different annexin domains in the protein sequence (Waseem et al. 2018).

CaAnn Gene Distribution on Chromosome

By using the coordinates in the PCD database, the location of different annexin sequences on chickpea chromosomes was determined.

Calculation of Ka/Ks for Gene Pairs

The Ka (nonsynonymous substitution rate) and Ks (synonymous substitution rate) values and their Ka/Ks ratios were calculated using TB Tool software (Plant et al. 2020). Then, the Ks values were used for the calculation of the approximate date of duplication events with the formula: $T = Ks/2r$, where r is the rate of divergence. For chickpea, assuming the clock-wise rate (r) of synonymous substitution of 6.05×10^{-9} substitutions per synonymous site per year (Jain et al. 2013).

Protein Sequence Alignment and Phylogenetic Analysis

The multiple sequence alignment of the CaAnn protein sequences was performed using CLUSTALW (<https://www.genome.jp/tools-bin/clustalw>), which had a gap opening penalty of 10.00 and a gap extension penalty of 0.05. For the phylogeny analysis between chickpea, *Arabidopsis thaliana* (*A. thaliana*), and *Medicago truncatula* (*M. truncatula*) annexin sequences, their protein sequences were used (Table S1). The phylogenetic tree was constructed by the neighbor-joining (NJ) method with MEGA 11 Software (Tamura et al. 2021). The bootstrap analysis with 1,000 replicates was used to increase the accuracy of the phylogenetic analysis. For better display, the various clades of Annexins were marked with different colors using the web server iTOL (Letunic and Bork 2021).

In Silico Analysis of Promoter Sequences of Chickpea Annexins

To understand better the annexin gene family, the cis-elements of the putative promoter sequences of chickpea annexins were analyzed. For this 1500 bp, upstream sequences from the start of the coding region of annexin genes were taken from PCD. The promoter sequences were analyzed by using the plant CARE web tool (Lescot et al. 2002) to identify cis-regulatory elements in the promoter sequences of chickpea annexins.

In Silico Gene Expression Analysis of Chickpea Annexins

For in silico expression analysis of annexins, transcriptome data were retrieved from the chickpea transcriptome database (CTDB) (<http://223.31.159.7/ctdb/>) (Garg et al. 2011; Singh et al. 2013; Verma et al. 2015a). Pfam ID for annexin was used to search annexin transcript identifier in the domain search tool of the CTDB. This showed fourteen transcript identifiers in the database. Then, the chickpea Annexin protein sequences were used as a query to find the relevant transcript and transcript identifier for them in the blast search tool of the CTDB. Transcripts with more than 99% identity with protein sequences were selected for expression analysis of ten CaAnn genes in five tissues, i.e., shoot, root, mature leaf, flower bud, and young pod, by using transcript identifiers as queries. The expression level of each CaAnn gene was presented using the RPM (reads per million) values. The heatmap of the expression data was generated by using the TB Tool software (Plant et al. 2020). Expression analysis of Chickpea was performed in drought stress by using transcriptome data at the early stage of development (Kudapa et al. 2023; Singh et al. 2023) and at early (flowering) and late (podding) reproductive stages (Garg et al. 2016) (<https://www.zhaolab.org/Legum-eIP/gdp/7/gene/profile/1>). Annexin expression data under desiccation, salinity and cold stress treatment of ten-day-old seedlings were also retrieved (Garg et al. 2015).

Results

Sequence Retrieval of Annexins in Chickpea

Thirteen non-redundant annexin encoding gene sequences (CaAnn1, CaAnn2, CaAnn3, CaAnn4, CaAnn5, CaAnn6, CaAnn7, CaAnn8, CaAnn9, CaAnn10, CaAnn11, CaAnn12, and CaAnn13) were identified in the genome sequence ICC 4958 v3.0, available in the PCD. Search in other databases like LIS and Phytozome showed similar results. Their respective mRNA, CDS, and protein sequences were also retrieved from the PCD (Table S2, Table S3 and Table S4, Table S5). The 13 Annexin genes that were identified encode proteins in the range

Table 1 Various features of *Cicer arietinum* Annexins

Gene Name	PCD ID	Gene Length	Chromosome Location	Start Position	End Position	Strand	CDS Length	Protein Length (aa)	mRNA Length	Annexin Repeats
CaAnn1	Ca_12411.1_v3	3063 bp	4	40,982,569	40,985,631	+	948 bp	316	1421 bp	4
CaAnn2	Ca_12411.2_v3	3244 bp	4	40,982,122	40,985,365	+	951 bp	317	1038 bp	4
CaAnn3	Ca_17735.1_v3	2499 bp	6	7,536,278	7,538,776	–	948 bp	316	1228 bp	4
CaAnn4	Ca_00608.1_v3	4010 bp	1	5,757,996	5,762,005	+	966 bp	322	1130 bp	4
CaAnn5	Ca_23918.1_v3	5745 bp	7	41,995,149	42,000,893	+	945 bp	315	1174 bp	3
CaAnn6	Ca_17171.1_v3	3314 bp	6	1,796,794	1,800,107	+	948 bp	316	1435 bp	4
CaAnn7	Ca_00606.1_v3	2595 bp	1	5,706,868	5,709,462	+	942 bp	314	1221 bp	4
CaAnn8	Ca_23921.1_v3	3783 bp	7	42,014,497	42,018,279	–	939 bp	313	1064 bp	4
CaAnn9	Ca_26832.1_v3	13121 bp	Scaffold02750	5278	18,398	+	948 bp	316	1237 bp	4
CaAnn10	Ca_00926.1_v3	2333 bp	1	9,558,572	9,560,904	–	942 bp	314	942 bp	4
CaAnn11	Ca_25972.1_v3	3157 bp	Scaffold00788	44,876	48,032	–	948 bp	316	1169 bp	4
CaAnn12	Ca_23919.1_v3	1285 bp	7	42,001,612	42,002,896	–	561 bp	187	561 bp	2
CaAnn13	Ca_28454.1_v3	818 bp	Scaffold14637	352	1169	+	219 bp	73	219 bp	1

of 73 to 322 amino acid residues (Table 1). The CaAnn annexin genes, mRNA, and CDS length range from 818 to 13121 bp, 219 bp to 1435 bp, and 219 bp to 966 bp respectively.

Identification of Conserved Motifs

Distributions of 20 motifs among the 13 chickpea Annexin proteins were studied using the MEME web server to examine the functions of CaAnn protein sequences (Fig. 1). 20 motifs varied in length from six to fifty amino acids and their number ranges from two to fifteen. Most proteins have eleven motifs, but 15, 12, 7, and 2 motifs were present in CaAnn5, CaAnn10, CaAnn12, and CaAnn13 respectively. The molecular function of these motifs was examined using Inter Pro Scan and Pfam database. The Pfam and InterProScan analysis indicated that only five motifs namely motifs 1, 2, 3, 4, and 6 were found to be associated with Calcium ion binding and Calcium-dependent phospholipid binding.

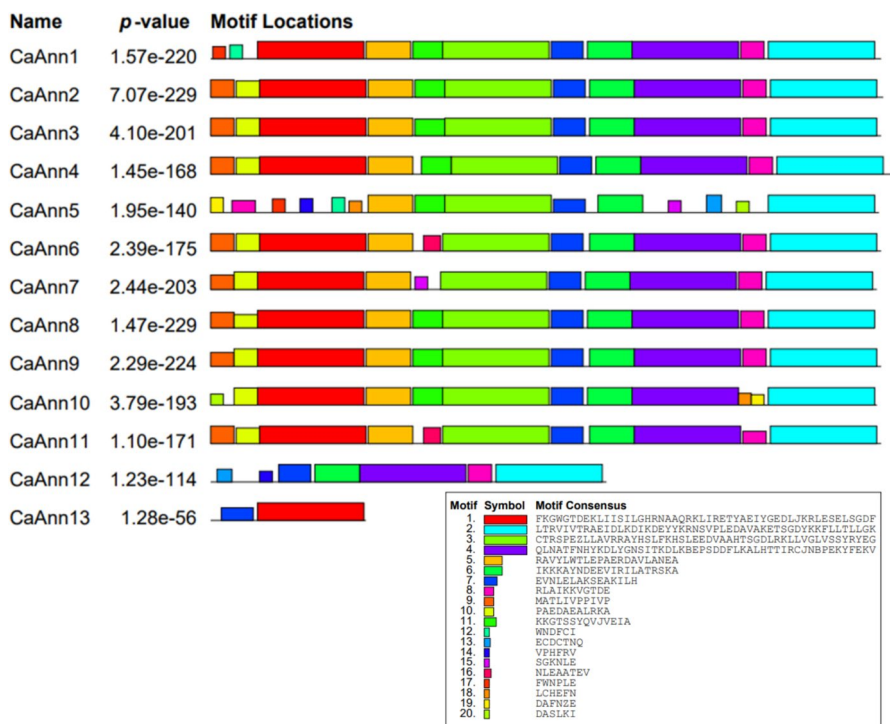


Fig. 1 Motif distributions of *Cicer arietinum* Annexin Proteins. The different colored boxes represent different motifs and their position in each annexin sequence. The sequences of key motifs are shown on the bottom (Color figure online)

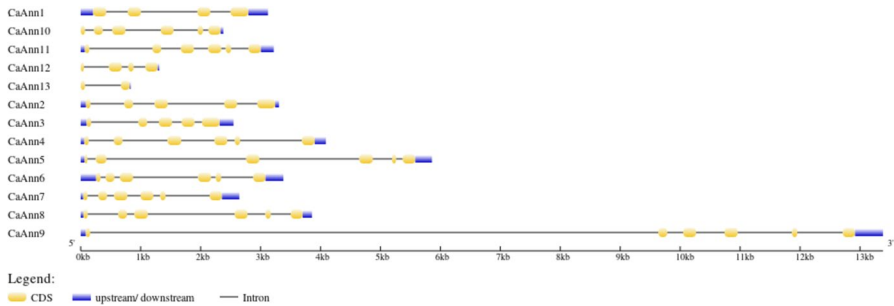
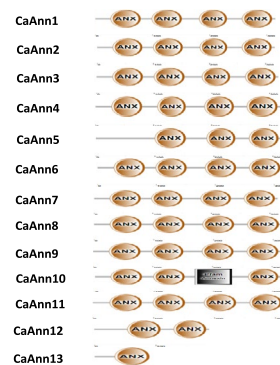


Fig. 2 Gene structure of *Cicer arietinum* annexins. The yellow box represents exons, blue box represents the untranslated region, and intron is represented by solid line (Color figure online)

Fig. 3 Analysis of conserved domains in thirteen chickpea annexins. The brown circles represent the presence of annexin domains (Color figure online)

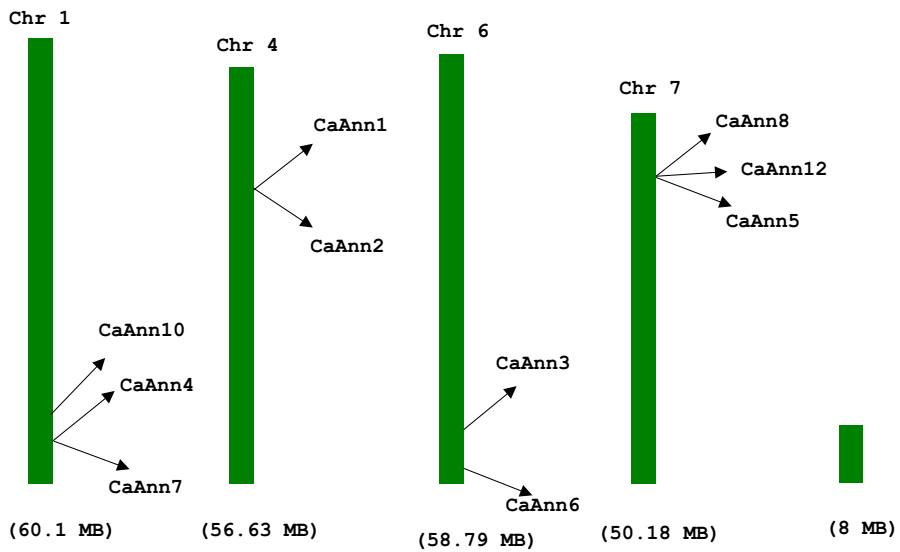


Prediction of Gene Structure

The online program Gene Structure Display Server (GSDS) 2.0 was used to analyze the exon/intron structure of the CaAnn gene to investigate its structural diversity. The result showed that in CaAnn genes the number of introns ranges from 3 to 5. CaAnn4, CaAnn5, CaAnn6, CaAnn7, CaAnn8, CaAnn9, CaAnn10, and CaAnn11 showed five introns' whereas CaAnn2 and CaAnn3 contained four introns and CaAnn1 and CaAnn12 contain three introns, while CaAnn13 showed the presence of only one intron (Fig. 2).

Analysis of Conserved Domains

The domain structure analysis showed that most of the CaAnn proteins possess four essential annexin domains (Fig. 3). Four conserved domains were found in ten annexin members while three annexin CaAnn5, CaAnn12, and CaAnn13, showed 3, 2, and 1 conserved domain, respectively. In CaAnn10, three annexin domains and one Pfam annexin domain were observed.



Gene Name	Chromosome	Start Position	End Position
CaAnn9	Scaffold02750	5278	18398
CaAnn11	Scaffold00788	44876	48032
CaAnn13	Scaffold14637	352	1169

Fig. 4 Chromosomal locations of *C. arietinum* annexins. Chromosome number is represented on the top of each chromosome and its size is represented in the brackets at the bottom of each chromosome. The 8 MB bar represents the scale for the four chromosomes (Color figure online)

Chromosomal Location of CaAnn Genes

The CaAnn gene sequences were used for the retrieval of the chromosomal location of CaAnn genes. The results showed that ten out of thirteen genes could be mapped on four chromosomes (Fig. 4). Chr 1 and Chr 7 have three genes on each i.e., CaAnn4, CaAnn7, and CaAnn10 were present in Chr 1. CaAnn5, CaAnn8, CaAnn12 were located on Chr 7. Chr 4 and Chr 6 showed two genes on each i.e., CaAnn1 and CaAnn2 are present in Chr4, whereas CaAnn3 and CaAnn6 were present in Chr 6. Two or more genes located on the same chromosome revealed the possibility of a tandem duplication event, while genes present on different chromosomes could be the result of a segmental duplication event. The high-density distribution of annexin genes on Chr 1 and Chr 7 strongly suggests the possibility of tandem duplication. Three genes i.e., CaAnn9, CaAnn11, and CaAnn13 were placed on the scaffold region.

Calculation of Ka/Ks for Paralogous Gene Pairs

To better understand the pressure of natural selection which is acting on paralogous CaAnn gene pairs, we calculated their Ka/Ks substitution ratios by TB Tool software. These indicators are also used to observe selection history during evolutionary processes and the expansion of gene families. Purifying selection is represented by a Ka/Ks value less than 1.00, neutral selection by a Ka/Ks value of 1, and positive selection by a Ka/Ks value greater than 1.00. In our study, we calculated the divergence period between the nine paralogous CaAnn gene pairs (Table 2). All the CaAnn paralogous gene pairs showed Ka/Ks ratios less than 1.00, which suggests the purifying selection of these nine paralogous gene pairs. Furthermore, these paralogous gene pairs' divergence times (in Ks values) ranged from 0.04 to 1.54. For the paralogous group CaAnn1/CaAnn2 and CaAnn8/CaAnn12, the estimated date of tandem gene duplication was calculated as 3.52 and 75.88 million years ago (MYA), respectively. While the segmental duplication time for the remaining seven paralogous pairs ranged from 63.45 MYA ($K_s = 0.7678$) for the paralogous group CaAnn9/CaAnn8 to 127.72 MYA ($K_s = 1.5454$) for CaAnn3/CaAnn1.

Protein Sequence Alignment of Chickpea Annexins

The identified CaAnn protein sequences were aligned using the CLUSTALW server, with default settings. The result of sequence alignment showed 93% similarity between CaAnn1 and CaAnn2, whereas CaAnn6 and CaAnn11 showed 76% similarity in their sequences. While CaAnn8 showed 75, 74, and 71% similarity with CaAnn9, CaAnn12, and CaAnn7 respectively. The conserved protein sequence (GXGT [38 residues]-D/E) determines the type II calcium-binding sites in annexins. The conserved sites were observed in the first and fourth repeats of the chickpea, but not in the second and third repeats (Fig. 5). In repeat I, GXGT Ca^{2+} binding sites were present and in repeat IV GTD Ca^{2+} binding sites were present. Putative S3 clusters which are involved in oxidative stress response were also present in all the chickpea annexins except CaAnn13. Besides, Ca^{2+} binding sites, the presence of Histidine (His) residue was also observed in the eleven CaAnn proteins. IRI

Table 2 Ka, Ks, and Ka/Ks values calculated for paralogous *Cicer arietinum* Annexin gene pairs

Sequence 1	Sequence 2	Ka	Ks	Ka/Ks	Date (Mya)	Duplicated Type
CaAnn1	CaAnn2	0.03163661	0.042699277	0.740916758	3.52	Tandem
CaAnn2	CaAnn3	0.215358715	1.536759358	0.140138216	127	Segmental
CaAnn3	CaAnn1	0.20487816	1.545486782	0.132565456	127.72	Segmental
CaAnn6	CaAnn11	0.13032527	0.955791113	0.136353298	78.99	Segmental
CaAnn7	CaAnn9	0.1930274	1.049154857	0.183983707	86.7	Segmental
CaAnn8	CaAnn12	0.18978514	0.918229821	0.206685881	75.88	Tandem
CaAnn9	CaAnn8	0.129368201	0.767810557	0.168489739	63.45	Segmental
CaAnn10	CaAnn13	0.353306072	1.467327422	0.240782028	121.26	Segmental
CaAnn12	CaAnn9	0.225791218	0.810941533	0.278430946	67.01	Segmental

Fig. 5 Protein sequence alignment of CaAnns. The alignment was performed using CLUSTALW. The putative annexin repeats i.e., I to IV are marked above the sequences. The conserved protein sequences are shown in different colors. The Red color indicates type II calcium-binding sites. The pink color indicates the putative S3 clusters. Dark yellow color indicates the RNSV conserve sequences. Histidine residues were highlighted with purple. Blue highlighted sequences show the IRI motifs. Dash indicates the sequence gaps to optimize the alignment (Color figure online)

conserved motifs were also found in nine chickpea annexins. Repeat IV showed the presence of RNSV conserved sequences in CaAnn1, 2, 3, 7, 8, 9 and 12.

Phylogenetic Relation of Chickpea Annexins with *Medicago truncatula* and *Arabidopsis thaliana* Annexins

To study the evolution of chickpea annexins, phylogenetic analysis was performed using Chickpea, *A. thaliana*, and *M. truncatula* Annexin protein sequences. The phylogenetic tree was constructed using a total of forty-one annexin protein sequences, including thirteen sequences from chickpea, twenty sequences from *M. truncatula*, and eight from *A. thaliana*. These annexins were clustered into three different groups (Fig. 6). Out of forty-one members, group I contained ten members, group II had twenty-one members, and group III had ten members. This phylogenetic distribution is conserved for different plant species, as group II comprised the highest number of annexin members, and groups I and III comprised the least members. Out of thirteen chickpea annexins, two members are present in group I, nine members in group II, and two members in group III. CaAnn6 showed 93% similarity with MtAnn9 and CaAnn5 showed 91% similarity with MtAnn4. CaAnn3 and CaAnn4 showed 89% similarity with MtAnn2 and MtAnn3, respectively. Most of the CaAnn genes formed clusters with MtAnn genes with high bootstrap values.

Calculation of Ka/Ks for Orthologous Gene Pairs

The divergence of orthologs between Annexin genes of chickpea and *M. truncatula* was analyzed by calculation of nonsynonymous substitution and synonymous substitution per site between different orthologous genes. A total of twelve pairs of orthologous genes were analyzed, and their Ka/Ks values were determined (Table 3). The result showed that all the twelve orthologous gene pairs had Ka/Ks ratios less than 1, indicating the predominance of purifying selection. The Ks value of ten orthologous pairs is less than 2, which indicates that these gene pairs were highly associated with duplication, whereas two orthologous gene pairs i.e., CaAnn13/MtAnn12 and CaAnn13/MtAnn13 showed a Ks value of more than two which could be associated with a high degree of uncertainty due to saturation of substitutions.

In Silico Analysis of Annexin Promoters

Using the PlantCARE database, cis-acting regulatory elements in the CaAnns promoter regions were predicted to study the transcriptional regulation of the CaAnn genes. The cis-acting regulatory elements present in CaAnn genes were further

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CaAnn1      EYQRKNSVPLDRAIVKDTISGDYKRMLLAILGHDDA--
CaAnn2      EYQRKNSVPLDRAIVKDTISGDYKRMLLAILGHDDA--
CaAnn3      EYQRKNSVPLDRAINDTISGDYKRMLLAILGHDDA--
CaAnn8      LYKRNNSVLEHEDVAREISGDYKFFLLTLGK-----
CaAnn12     LYKRNNSVLEHEDVAREISGNYKFFLLTLGKHE-----
CaAnn9      LYKRNNSVLEHEDVAREISGDYKFFLLTLGKQD-----
CaAnn7      VYKRNNSVLEHVIAREISGDYKFFLLTLGKED-----
CaAnn13     -----
CaAnn10     LFVKRNNSVTHDITVARTWGNVYKFFLLLELGG-----
CaAnn6      EYLRKYKRTLLDVAHSESTGYSKRAFLSLGNGNQ----
CaAnn11     EYLRKYKRTLLDVAHSESTGYSKRAFLSLGNGNQ----
CaAnn4      EYANNMYKSLDDQDIETGAKNGYKFFLLTLGKPKPGD--
CaAnn5      EYNNLYKGSVLDDQDIETGAKNGYKFFLLTLGIARG---

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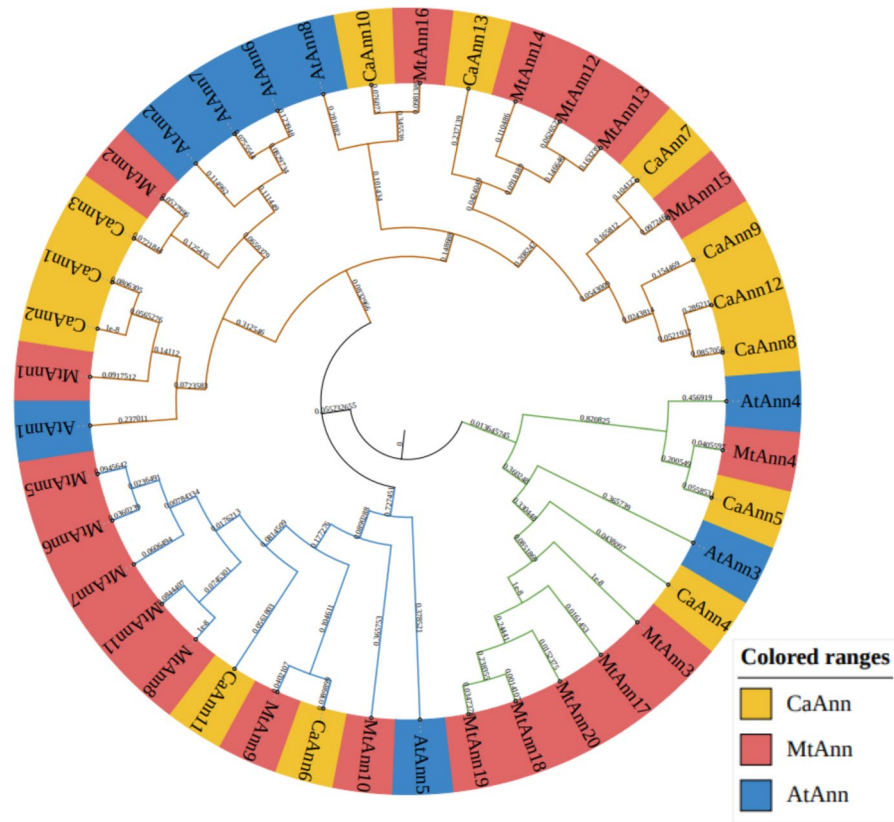


Fig. 6 Phylogenetic tree showing the similarity between *Cicer arietinum*, *Medicago truncatula*, and *Arabidopsis thaliana* Annexins. Different groups highlighted with different colors. And different protein sequences are marked with three different colors (Color figure online)

Table 3 Ka, Ks, and Ka/Ks values calculated for orthologous *Cicer arietinum* and *Medicago truncatula* Annexin gene pairs

Sequence 1	Sequence 2	Ka	Ks	Ka/Ks	Date (Mya)
CaAnn1	MtAnn1	0.101070878	0.29306495	0.344875354	24.22
CaAnn2	MtAnn1	0.070589491	0.250543676	0.281745253	20.7
CaAnn3	MtAnn2	0.060467481	0.318204025	0.19002739	26.29
CaAnn4	MtAnn3	0.057996524	0.46046405	0.125952338	38.05
CaAnn5	MtAnn4	0.046206141	0.218706119	0.211270453	18.07
CaAnn6	MtAnn9	0.0364109	0.342952713	0.106168865	28.34
CaAnn7	MtAnn15	0.100226951	0.593911516	0.168757379	49.08
CaAnn10	MtAnn16	0.076356446	0.258334565	0.295571932	21.34
CaAnn11	MtAnn11	0.110799477	0.585805526	0.189140375	48.41
CaAnn13	MtAnn12	0.846177389	3.501812634	0.241639824	289.4
CaAnn13	MtAnn13	0.83926587	2.499658603	0.335752198	206.58
CaAnn13	MtAnn14	0.218438825	0.872889608	0.250247938	72.13

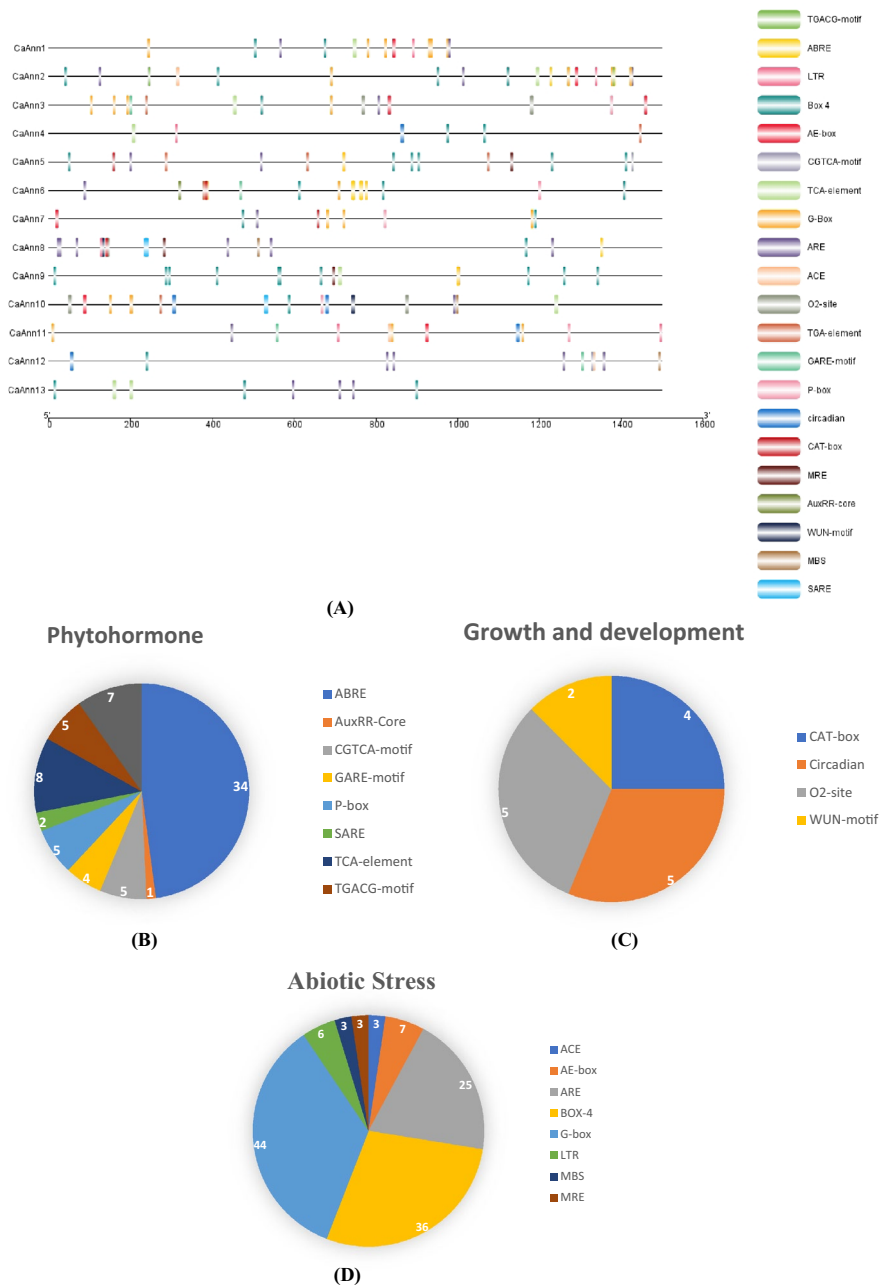


Fig. 7 Cis- acting regulatory element in the promoter regions of CaAnns. **A** Distribution of cis-acting elements in each CaAnn promoter. **B** The frequency distribution chart of phytohormone responsive element. **C** Distribution of growth and development responsive element. **D** Abiotic stress-responsive element in CaAnn promoter. Different elements are indicated by different colors in the grid (Color figure online)

classified into three types i.e., phytohormone responsive element, abiotic stress-responsive element, and growth and development response element (Fig. 7A). Phytohormone responsive elements including ABRE, AuxRR-Core, CGTCA-motif, GARE-motif, P-box, SARE, TCA-element, TGACG-motif, and TGA-element were present (Fig. 7B). Abiotic stress-responsive elements including ACE, AE-box, ARE, BOX4, G-box, LTR, MBS, and MRE were present (Fig. 7D). Growth and development responsive elements including CAT-box, Circadian, O₂-site, and WUN-motif were found in the CaAnn promoter sequences (Fig. 7C). Two cis-responsive elements, i.e., Box4 and ARE, were found to be present in all annexins.

Annexins Expression in Chickpea

The gene expression data from CTDB for different chickpea annexins were used to understand the expression pattern in different tissues (Table S6). The RPM values for annexin CaAnn1, CaAnn2, CaAnn3, CaAnn4, CaAnn5, CaAnn6, CaAnn7, CaAnn8, CaAnn9, and CaAnn11 in different tissues like root, shoot, mature leaf, flower bud, and young pods showed differential expression pattern. In the CaAnn1 and CaAnn2 transcripts, the expression level was higher in Young Pod, i.e., 150.1 RPM, as compared to other tissues (Fig. 8). The transcripts of CaAnn1 and CaAnn2 were similar because both are represented by the same transcript. In CaAnn3, the expression was relatively higher in young pods as compared to the expression in flower bud, root, and shoot, respectively, and no expression was observed in mature leaves. In comparison to other annexins, CaAnn4 showed relatively low expression levels in the root and young pod. In CaAnn5 and CaAnn8, the expression was observed only in the root and young pod, respectively. In CaAnn6, the expression in the root was relatively high as compared to other tissues. In CaAnn7, the expression was relatively high in the young pod, then decreased sequentially in the flower bud, root, shoot, and then in mature leaf, respectively. CaAnn9 showed relatively high expression in young pods as compared to other tissues, whereas no expression was observed in mature leaves. CaAnn11 showed relatively high expression in the root in comparison to other tissues. The expression of CaAnn1, CaAnn2, CaAnn6, CaAnn7, and CaAnn11 was observed in all tissues. In vegetative organs, the expression of CaAnns was relatively high in roots except for CaAnn9, which showed relatively higher expression in the shoot. The mature leaf showed a relatively low expression level in all annexins compared to the root and shoot except CaAnn11, which showed relatively low expression in the shoot. In reproductive organs, the young pod showed relatively high expression as compared to the flower bud in all annexins except CaAnn11, which had higher expression in the flower bud than that of the young pod. CaAnn7 was strongly expressed in the young pod. The ICC 4958 showed different root traits and have been characterized extensively for their contrasting responses to drought stress. During early plant development under drought conditions, only CaAnn8 showed significant downregulation of expression level (Fig. 9A) Table S7. However, during early (flowering), CaAnn7 and CaAnn10 show significant downregulation and upregulation, respectively, while

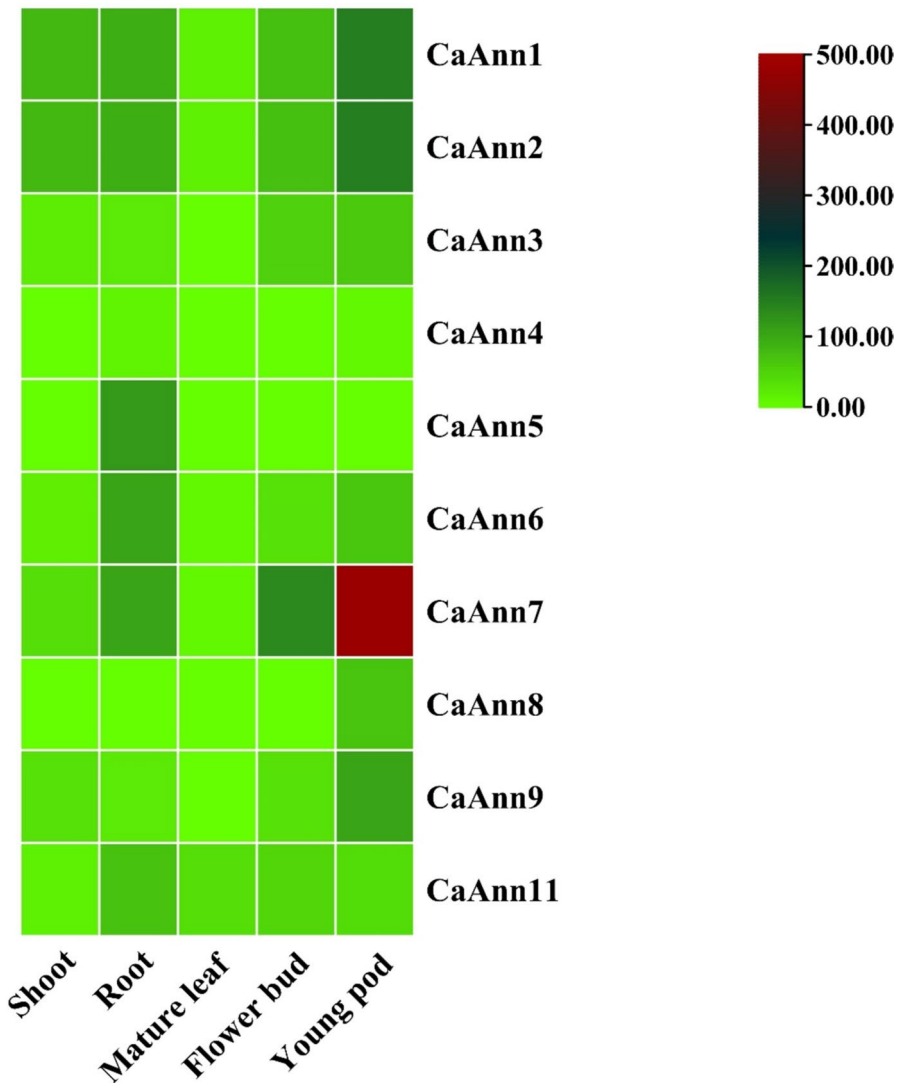
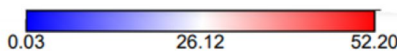


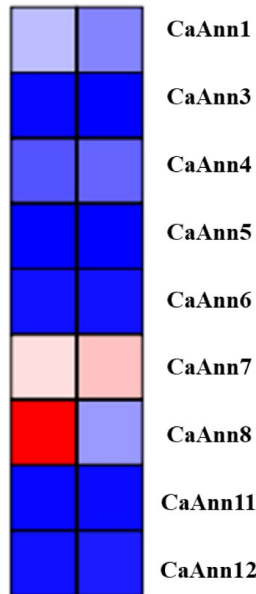
Fig. 8 This heatmap represents shoot and root expression data for 15-day-old seedling. The expression data for mature leaf, flower bud, and young pod were obtained from field grown plant. For each tissue, three biological replicates were used for tissue-specific expression analysis (Garg et al. 2011). The color scale (0–500) bar represents the RPM values as mentioned in CTDB database for different genes. Light green indicates relatively low expression, dark green indicates relatively moderate expression, and brown indicates relatively high expression (Color figure online)

during late (podding) reproductive stages, no significant changes were observed (Fig. 9B) Table S8. The transcriptome data retrieved from (Kudapa et al. 2023) showed differential expression of chickpea annexin in drought (Table S9). The annexin expression data in root and shoot under desiccation, salinity, and cold stress were also analyzed (Garg et al. 2015) Table S10.

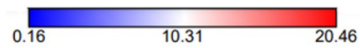
(A)



Control
Drought



(B)



50 d Control
50 d Drought
70 d Control
70 d Drought

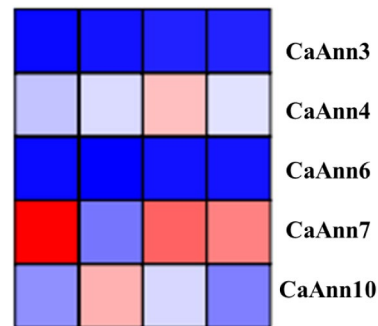


Fig. 9 Heatmap representing CaAnn gene expression under drought conditions in root tissue at different stages of development. **A** Expression profile for plant root tissue with three biological replicates after withholding water for 15-day in 11-day-old plants (Singh et al. 2023) and **B** Expression profiling for root tissue with three biological replicates at early (flowering, 50 days) and late (podding, 70 days) reproductive stages (Garg et al. 2016). Heatmap was plotted using FPKM values (Color figure online)

Discussion

Plant cells respond to biotic or abiotic stress by changes in the intracellular pattern of calcium which sets off a signaling cascade. Plant Annexins function as major calcium sensors and control a variety of targets in cellular signaling networks (Laohavisit et al. 2010; Zhao et al. 2023). The identification and characterization of Annexin family members in protein-rich food legume crops like chickpea provide chances for the functional validation of these members in the abiotic stress. This study carried out a detailed genome-wide investigation, which included annexin gene identification in chickpea genome, their phylogenetic relationships study, evolutionary

analysis, conserved domain and motifs prediction, chromosomal positions of different chickpea annexins, analysis of their gene structure, identification of cis-elements in their promoters. Moreover, the gene expression of CaAnn was also analyzed in various tissues under different stages of growth and development. How different abiotic stress conditions regulate the annexin expression in chickpea was also analyzed by available transcriptome data.

A thorough analysis of several databases resulted in the identification of thirteen non-redundant Annexin genes in the Chickpea genome, annexin family reported in other plant species, including eight in *Arabidopsis thaliana* (Cantero et al. 2006), nine in *Solanum lycopersicum* (Lu et al. 2012), ten in *Oryza sativa* (Jami et al. 2012a), twelve in *Zea mays* (Zhou et al. 2013), thirteen in *Brassica rapa* (Yadav et al. 2015; He et al. 2020), five in *Brassica juncea* (Jami et al. 2009), and eleven in *Hordeum vulgare* (Xu et al. 2016). The variation in the number of Annexin genes among various species may be due to the natural variation among those species and their adaptability to the challenging growth environment. The function of Annexin members partly depends upon the number of motifs involved in Ca^{2+} binding. In Chickpea annexin, 2 to 15 conserve motifs were found. Some motifs, such as 1, 2, 3, 4, and 6, which are Ca^{2+} ion binding and Ca^{2+} -dependent phospholipid-binding motifs, were related to the annexin function. The exon–intron pattern shows the structural diversity of CaAnn genes (Shen et al. 2021). It is assumed that, if a low number of introns are present in a gene, then the plants will respond more quickly to environmental changes (Jeffares and Penkett 2000). Here, CaAnn13 could respond better to environmental changes because it contains only one intron. The domain structure analysis showed that out of thirteen annexin members, ten members contain four essential annexin domains. Among them, CaAnn10 contain 3 annexin domains and one Pfam annexin domain. The annexin domain refers to the specific protein structure involved in essential cellular processes. In contrast, the Pfam annexin domain defines the classification and identification of the annexin domain within the Pfam database, which serves as a resource for comparative genomics and protein function research. The results of gene location on chromosomes showed that ten out of thirteen genes could be mapped on four chromosomes and three genes are present on the scaffold region. The plant genome has expanded significantly through tandem and segmental duplication events (Sun et al. 2015). Among CaAnn paralogous gene pairs, seven pairs were originated from segmental duplication and two pairs in tandem duplication events (Table 2). In the Annexin gene family, segmental duplication is a common expansion mechanism. The primary mechanisms that cause the expansion of gene families are tandem duplications and segmental duplications (Zhang et al. 2015). On the chromosomes of *C. arietinum*, they all formed gene clusters for tandemly duplicated gene pairs. We also observed that there were numerous orthologous gene pairs in the CaAnn and MtAnn genes, which could be due to the close relation between *C. arietinum* and *M. truncatula* based on the phylogenetic analysis (Ling et al. 2020). The prediction of the physiological function of genes using phylogenetic analysis can potentially be helpful (Faik et al. 2006). Similar to the earlier reports of rice annexin by Jami et al 2012a, the presence of IRI motifs in some chickpea annexins also showed their involvement in F-actin and GTP-binding regions for phosphodiesterase activity (Jami et al. 2012a). Analysis of the selection

pressure in gene duplication occurrences is based on the substitution rates of Ka and Ks (Ivanova et al. 2017). We identified that Ka/Ks values of all the orthologous and paralogous gene pairs were less than one, indicating that they had usually evolved under the purifying selection (Die et al. 2018; Song et al. 2020; Moinoddini et al. 2023). The phylogenetic analysis between *Cicer arietinum*, *Medicago truncatula*, and *Arabidopsis thaliana* was observed to see the evolutionary relationship within the same family and between two different families. The *Cicer arietinum* and *M. truncatula* belong to the same family Fabaceae and *Arabidopsis thaliana* is quietly related to chickpea, which belongs to the Brassicaceae family. The 41 annexin members from *Cicer arietinum*, *M. truncatula*, and *Arabidopsis thaliana* were clustered into three groups. It also revealed the presence of several orthologs, known to be originated by splitting of taxonomic lineages and retaining similar functions and paralogs, assumed to be created by gene duplication events possessing different functions. In the Phylogenetic tree if the two genes are very similar and grouped together with high bootstrap value indicates that they are results of gene duplication. In our study from the phylogenetic tree, within the separate clades, Chickpea annexins were closer to the *M. truncatula* annexins which could be because of *M. truncatula*, and chickpea comes under the same family, whereas distantly placed from *Arabidopsis thaliana*. Here, AtAnn1 is orthologous to CaAnn1 and CaAnn2, which suggests that the CaAnn proteins clustered with same group can possess similar biological functions.

In this study, we observed that each CaAnn gene promoter sequence possessed at least one cis-element related to stress response, growth and development-related response, and hormone response. The presence and distribution of various cis-regulatory elements affect the spatial and temporal expression of genes. The predicted gene functions could be examined using bioinformatics-based evaluation of cis-regulatory regions (Nasim et al. 2016). The cis-elements are significant molecular switches that control how genes are expressed in response to abiotic stress, and they can be activated by both ABA-dependent and ABA-independent signal transduction pathways. In silico analysis showed that except CaAnn10, 12, and 13, each chickpea annexin possesses ABRE elements in its promoter sequence. The ABRE motif is activated during salt, dehydration, cold, and heat stress and follows a pathway of ABA-dependent and independent signal transduction mechanism (Jami et al. 2012a). MBS and MRE are described as an MYB binding site, which is involved in drought-inducibility and salinity. 11 out of 13 CaAnn promoters contain the MYB motif (Haider et al. 2019; Khan et al. 2019). CaAnn promoters with components like MYB and ABRE may be involved in controlling the development of chickpea plants under abiotic stress (Deepika et al. 2022). The plant system's circadian rhythm is regulated by the circadian cis-regulatory element. LTR is a cis-acting element that is well known for controlling processes under harsh conditions like dehydration, cold, and salt stress (Shen et al. 2021). Few cis-regulatory components, such as ethylene, auxin, gibberellin, and salicylic acid, are actively responsive to hormonal stressors (Khan et al. 2019). These results show that CaAnn genes might play critical roles not only in plant growth and development response but also in various phytohormone-mediated abiotic stress responses in Chickpea.

The expression of plant annexins in different tissues seems to be related to growth and development in different plants (Yadav et al. 2018). In this study, CaAnn genes showed differential expression patterns in different tissues, suggesting their specific or significant functions. The ability to study the *in silico* expression of CaAnn genes in different tissues has been made possible by the availability of tissue-specific transcriptome data in chickpea. Previous studies have shown the expression of annexins from different plants and in different tissues (Yadav et al. 2018). In *Arabidopsis thaliana*, the expression of AtAnn1 showed their role in mature and germinated pollen, root development, and growth (Noir et al. 2005; Zou et al. 2009; Laohavisit et al. 2012; Wang et al. 2018). CaAnn1 and CaAnn2 were homologous to AtAnn1, suggesting they might have a role in root growth and in mature and germinated pollen. AtAnn5 showed their expression in pollen germination, pollen development, and pollen tube growth (Zhu et al. 2014). CaAnn6 shares an identity with AtAnn5, which suggests it might be involved in embryo and pollen formation. In *Triticum aestivum*, TaAnn1 and TaAnn2 showed preferential expression in the root (Xu et al. 2016). In *Brassica juncea*, AnnBj2 showed their role in root growth (Ahmed et al. 2017). In *Gossypium barbadense*, the expression of GbAnn6 showed their role in increased root cell length (Huang et al. 2013). The *M. truncatula* annexin showed the expression of MtAnn2 in arbuscule-containing cells of mycorrhizal roots (Manthey et al. 2004). CaAnn3 is orthologous to MtAnn2 and AtAnn2, so it might be involved in root development (Wang et al. 2018). In *Phaseolus vulgaris*, the transcript of PvAnn1 showed the highest accumulation in the root (Carrasco-Castilla et al. 2018). In the *Solanum lycopersicum* plant, where AnnSl1 showed the expression in mature and germinated pollen (Lu et al. 2012). *Arachis hypogaea* annexin showed a strong expression of AnnAh3 and AnnAh5 in flowers (He et al. 2015). In CaAnn7, the expression was relatively high in flower bud and young pod, which shows its possible involvement in reproductive stages. A gene's transcriptional regulation depends on the cis-regulatory elements present in the promoter (Hernandez-garcia and Finer 2014). Four major cis-acting regulatory elements i.e., CAT-box, circadian, O2 site, and WUN motif were observed among different chickpea annexins. The role of CAT-box is for meristem expression, suggesting CaAnn plays a role in the root, growth of the stem, and flower development (Yihong et al. 2016). The cis-acting regulatory element, circadian was involved in circadian control. Role of the O2 site is found in zein metabolism regulation and wound motif is responsible for wound responsiveness. AnnBj1 is homologous to AnnAt1, involved in wound signaling (Jami et al. 2008).

Transcriptomic analysis of chickpeas (ICC4958) under drought stress showed that nine genes in 11-day-old plants and five genes in early (flowering, 50 days) and late (podding, 70 days) reproductive stages were differentially expressed in the root of chickpeas (Garg et al. 2016; Singh et al. 2023). In *Arabidopsis thaliana*, AtAnn8 positively conferred drought stress tolerance, but AtAnn1 and AtAnn4 showed negative regulation during drought stress (Huh et al. 2010; Yadav et al. 2016). CaAnn1 shares an identity with AtAnn1 and its expression is negatively regulated under drought stress. In case of CaAnn10 which is homologous to AtAnn8, shows its positive regulation in the 50-day-old stage, while it was negatively regulated in the 70-day-old stage. The expression data of

short-term exposure to cold stress of the seedling stage showed positive regulation of CaAnn7 in shoot and negative regulation in the root (TableS6). While in salinity stress it showed downregulation in shoot and in upregulation in roots. This different response under different stress conditions of same annexin in chickpea could be due to their tissue-specific and stage-specific expression under stress (TableS6).

Our study identifies and describes thirteen annexin members in chickpea (*Cicer arietinum*), focusing on its possible function in response to stress. Chickpea is an essential legume crop grown particularly in semi-arid and arid regions where abiotic factors like salinity and drought significantly reduce crop productivity (Kumar and Abbo 2001). By identifying different annexin genes that are differentially expressed in different stress conditions, our study helps in the possible targets for improvement strategies that aim to improve the stress tolerance in chickpea. The annexin is a calcium-binding protein known to play a role in different cellular processes such as in signaling cascades, membrane repair, and cytoskeletal organization. These essential cellular processes are utilized in cellular homeostasis and allow the plants to cope with environmental challenges, eventually improving growth and productivity. By the integration of our findings with the existing knowledge of annexin gene function and utilization of different biotechnology techniques and modern breeding programs, our study provides significant improvement in agricultural productivity.

Conclusion

Genome-wide identification and in silico analysis of plant annexins have been widely studied in diverse plant species. However, a genome-wide study of the annexin gene in essential legume crop chickpea was missing. Therefore, our study focused on genome-wide identification and analysis of the annexin gene family in chickpea. A total of thirteen non-redundant annexin genes were identified in *C. arietinum* genome. Phylogenetic analysis and protein structure showed that the annexin family in chickpeas is highly conserved based on evolution and structure. Based on in silico expression studies carried out at different developmental stages, annexins are expressed in a variety of tissues and could be essential for their role in the growth and development of plants. The promoter and expression analysis of the annexin gene family shows their involvement in abiotic stress response. Importantly, some of the members of CaAnn genes might respond to drought stress. In the future, this study can provide a helpful platform for detailed functional characterization of the annexin family in chickpea and comparative analysis in other legume crops.

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Author Contributions D.Y. designed the experiments. B.S. and P.G. performed the experiments. B.S., P.G. and D.Y. analyzed the data and wrote the manuscript.

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Declarations

Competing Interests The authors declare no competing interests.

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